Inspectable Clinical Extraction from Epilepsy Letters

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Abstract—**Clinical extraction results are difficult to interpret when model output, deterministic rules, repair, evidence checks, and scoring are reported as one method. We evaluate three methods on two epilepsy-letter tasks: current seizure-frequency labeling on the Gan 2026 synthetic benchmark and broad epilepsy phenotyping on ExECTv2. For each task we compare rules only, LLM only, and LLM with rules. Saved Gan holdout results are 364/450 Purist for the single-pass system and 379/450 for a multi-model comparison. On ExECT dev140, rules only reached 0.6020 paper-derived all-features macro item F1 (0.3548 under the existing strict micro scorer), the LLM only method reached 0.7393 clinical fact F1, and the combined method reached 0.9189. The fixed six-model comparison is incomplete. The paper-facing table retains completed historical GPT and Qwen rows; DeepSeek V4 Flash is pending because only a thinking-enabled result will be reported. Saved-output ablations show positive normalization contributions on both tasks, while the evidence check is score-inert on those representative replays. The selected evidence supports a reproducible component comparison with explicit data limits and a tested implementation of the published ExECT metric views. It does not reproduce the original ExECT system or its reported 0.87/0.90 scores, and it does not support independent clinical validation or a completed six-model result.**

Index Terms—**clinical information extraction, epilepsy, large language models, reproducibility, evidence attribution**

I. Introduction

Epilepsy letters combine temporal reasoning, clinical terminology, medication regimens, investigation findings, and ambiguous current-versus-historical statements. Large language models (LLMs) can broaden extraction, but a model call can hide where a result came from. Deterministic systems are easier to inspect, yet often fail on linguistic variation and long-range clinical selection.

We compare rules-only, LLM-only, and LLM-with-rules methods. The same package contains a deep single-label task and a broad multi-entity task. This makes it possible to ask which stages transfer, which results depend on task-specific scoring, and where evidence is insufficient for a stronger claim.

The paper contributes: (1) selected results for three methods on two tasks; (2) explicit ownership of extraction, normalization, final formatting, repair, evidence checking, and scoring; (3) saved aggregate Gan holdout results and partial ExECT model-transfer evidence; and (4) reproducibility controls that pin prompts, scorers, splits, repair policies, runtime identifiers, dependencies, and file hashes.

II. Related Work

ExECT demonstrated structured epilepsy extraction from unstructured clinic letters and later work documented the corresponding annotation resource [1], [2]. Seizure-frequency extraction has also been studied with machine-reading and pretrained-language-model approaches [3], [4], while recent work has explored fine-tuned LLMs and reproducible synthetic clinical letters [5], [6]. Our focus is complementary: processing steps and data limits are part of the scientific method, and rules-only, LLM-only, and LLM-with-rules methods are compared without collapsing their different scoring and repair mechanisms.

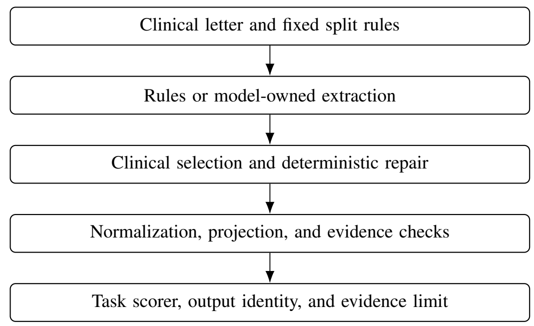

Fig. 1. Processing steps and their owners. Ownership remains explicit when a step is implemented by an LLM.

III. Methods

A. Tasks and Split Policy

Gan 2026 asks for one current seizure-frequency label per synthetic letter [6]. Validation750 is the row-inspectable development and replay split. Test450 is a locked holdout: only saved aggregate results are retained, and row-level holdout failures are not used for development.

ExECTv2 covers Diagnosis, SeizureFrequency, Prescription, and Investigations in the primary comparison, with additional entities retained in the deterministic all-nine reference [1], [2]. Dev140 is the row-inspectable development split. Full200 combines dev140 with held-out test60 and is reported only as a development-inclusive aggregate audit; it is not an independent holdout.

B. Compared Methods

For each task, the retained evidence index selects one rules-only, LLM-only, and LLM-with-rules run. Rules-only runs use no model calls. LLM-only runs put the clinical decision in one model program. The ExECT GEPA run is a negative development comparison. Combined runs keep model extraction distinct from deterministic selection, normalization, evidence checks, and scoring.

TABLE I
SELECTED RESULTS FOR THREE METHODS ON TWO TASKS

Task	Method	Split	Selected result	Use and limit
ExECTv2	Rules only	dev140	All-feature macro item F1 0.6020	Paper-derived development result; strict micro F1 0.3548.
ExECTv2	LLM only	dev140	Headline F1 0.7393	GEPA negative development comparator.
ExECTv2	LLM with rules	dev140	Clinical fact F1 0.9189	Current development reference.
Gan 2026	Rules only	validation750	697/750 Purist	Deterministic development comparator.
Gan 2026	LLM only	validation750	581/750 Purist	Single-pass development comparator.
Gan 2026	LLM with rules	validation750	661/748 rendered Purist	Single-pass event comparison.

C. *Scoring, Repair, and Attribution*

Gan uses Purist and Pragmatic label scoring; Purist is the primary strict label score. ExECT's primary internal score is de-duplicated `clinical_headline` recovery across four families. Phrase, CUI, evidence-valid, and full-attribute companions are reported separately. `clinical_headline` is not presented as reproduction of the published strict ExECT benchmark. The paper-derived views score each entity type separately and report their macro mean. Per-item scoring counts mentions; per-letter scoring requires at least one correct mention and feature bundle. Certainty applies to Diagnosis and PatientHistory; negation applies only to PatientHistory.

Raw model selection, JSON or format repair, source-exact evidence repair, deterministic semantic repair, projection, and final scoring are separate stages. Model-specific transport or output-shape adapters do not become model quality evidence. Semantic repairs retain a named deterministic owner and must be ablated or otherwise tested before supporting a causal interpretation.

D. *Recorded Reference Version*

The retained evidence index records implementation commit `46562134`, the dependency declaration and lock, prompts, scorers, split definitions, repair policies, model policy, split runbook, and CI workflow. A semantic prompt, scorer, split, repair, runtime-route, or component-graph change requires a new recorded version and complete replay. This rule does not authorize model calls.

IV. RESULTS

A. *Selected Method Comparison*

Table I is the minimum selected scientific comparison. Scores are not compared across tasks because the datasets, outputs, and scorers differ.

All six runs replay from current code and selected outputs without model calls. The ExECT combined-method replay also returns evidence-valid F1 0.8913. Its legacy benchmark/CUI companion is 0.4791 versus 0.4729 in the saved run; this diagnostic does not define the paper-derived rules-only result or affect the reproduced 0.9189 headline score.

TABLE II
EXECT RULES-ONLY DEV140 METRIC VIEWS

View	Per-item F1	Per-letter F1
Normalized phrase	0.5687	0.7518
CUI	0.7144	0.8534
All features	0.6020	0.7922

TABLE III
GAN SAVED RESULTS ON LOCKED TEST450

System	Purist	Role
Single structured-event pass	364/450	One model pass per note
V12 multi-trace reviewer	379/450	Three model passes per note

TABLE IV
EXECT SAME-CORE FULL200 AGGREGATE `CLINICAL_HEADLINE`

Model condition	Overall	Diagnosis	SF	Prescription	Investigations	Call/parse failures
DeepSeek V4 Flash (thinking)	Pending	Pending	Pending	Pending	Pending	Pending
GPT-4.1-mini	0.8356	0.8397	0.7525	0.8926	0.8563	0/0
Qwen 3.6 35B repair v02	0.8197	0.8307	0.7020	0.8926	0.8503	0/0

B. *ExECT Paper-Derived Metric Replay*

The no-call replay covers all nine entity types. CUI identity recovers many surface-form misses, while attribute agreement—especially for Diagnosis—is the main remaining loss. One gold mention and six predictions lack a CUI; missing identifiers never match. The original 0.87 per-item and 0.90 per-letter scores remain reference values, not reproduced results.

C. *Gan Locked Holdout Evidence*

Both rows in Table III are saved aggregate evidence. The 15-row quality difference is 3.33 percentage points. A cold execution requires one model pass per note for the single-pass system and three for the saved V12 test condition. DeepSeek input was unavailable on all 450 rows. The saved V12 audit made 450 new reasoner calls while replaying two upstream traces. Matched tokens, cost, latency, hardware, and cache telemetry were not retained, so no measured token, dollar, energy, or latency comparison is claimed.

D. *ExECT Same-Core Model Evidence*

The retained model-transfer package holds the two-call no-SF-adjudicator component graph fixed while changing the runtime model. Table IV is a full200 development-inclusive aggregate read; no test60 row-level analysis is used.

The completed results apply to the two named historical runtime conditions. The DeepSeek V4 Flash condition uses the `deepseek/deepseek-chat` API identifier with thinking enabled; only that condition will be reported, and its result is pending. The completed rows are not a strict same-prompt panel: the retained GPT condition used temperature 0.3, and Qwen used a compact prompt and output-contract repair. The requested six-model comparison remains incomplete. The fixed roster is GPT-4.1-mini, GPT-5.6 Luna, GPT-5.6 Sol, DeepSeek V4 Flash, Qwen 3.6:35B, and Gemma 4 26B.

TABLE V
SAVED-OUTPUT COMPONENT ABLATION

Component	ExECT dev140	Gan validation750
Normalization/dictionary	+0.0389	+0.0293
Evidence validation	0.0000	0.0000

TABLE VI
RETAINED RELIABILITY EVIDENCE AND LIMITS

Subject	Retained result	Boundary
ExECT evidence	Hybrid dev140 minimum exact evidence rate 1.0000	Development reference.
ExECT calibration	Brier 0.2225; base-rate Brier 0.2340; ECE 0.0587	Full200 aggregate internal scoring-rule result.
Gan grounding	Architecture-specific validation grounding packages retained	Metrics are not uniform across families.
Reproducibility	1,157 tests, Ruff, mypy, hashes, split restrictions, and six-run replay pass on Python 3.11	Engineering verification, not clinical validation.

E. Cross-Task Components and Reliability

The zero score delta in Table V does not show that evidence validation is unnecessary. A filter may be essential on malformed or hallucinated outputs even when all selected reference predictions already satisfy it. Rejection and repair tests, not aggregate F1 alone, carry that part of the evidence.

Model-reported confidence remains unused, and no low-burden review-routing policy is adopted. The retained evidence index does not include evidence for a cross-task over-reading claim.

V. DISCUSSION

The reduced system can reproduce six selected runs while keeping component ownership and data limits explicit. The combined ExECT method is substantially stronger than the retained ExECT rules-only and LLM-only development references on `clinical_headline`, but the three runs do not share the paper-derived metric views and cannot establish strict benchmark superiority. The rules-only replay shows that CUI matching recovers surface variation, while feature completion remains the larger limitation.

The historical model-transfer artifacts show that the component graph ran with three qualitatively different runtime models. The old DeepSeek aggregate is excluded from the paper-facing comparison because its thinking state was not recorded. Because the completed runtime conditions are asymmetric and the fixed six-model panel is incomplete, these results do not support a universal model claim.

The cross-task ablation suggests that normalization is useful in both tasks. The exact-evidence check's zero F1 delta shows why aggregate score changes are insufficient: rejection, source attribution, and repair need direct tests.

VI. LIMITATIONS

Gan test450 contains saved aggregate evidence only and cannot support row-level tuning. ExECT full200 includes dev140 and is not an independent holdout. `clinical_headline` is a clinical fact score, not the published strict benchmark. The paper-derived replay uses development data and does not reproduce the original system, annotation process, or 0.87/0.90 scores. A legacy benchmark/CUI companion has small current-code replay drift. The fixed six-model panel is incomplete and the completed rows use asymmetric runtime conditions. Model-reported confidence and low-burden review routing are not validated. Annotation-quality findings are internally adjudicated; unqualified clinical validity would require independent domain review. The current retained evidence does not support a cross-task over-reading claim.

VII. CONCLUSION

An inspectable clinical extraction system can preserve performance evidence and the source needed to interpret it. The selected results support three methods on two tasks, saved Gan holdout results, and partial ExECT model runs. They do not support strict ExECT score reproduction, a six-model conclusion, or independent clinical validation. Those boundaries are part of the result, not footnotes to it.

REPRODUCIBILITY STATEMENT

The machine-readable retained evidence index records the six runs, selected files and hashes, policy versions, and no-call replay expectations. The Markdown manuscript and this IEEE source use only selected evidence and claims permitted by the paper claim status.

REFERENCES

[1] B. Fonferko-Shadrach, A. S. Lacey, A. Roberts, S. Thompson, D. V. Ford, R. A. Lyons, M. I. Rees, and W. O. Pickrell, "Using natural language processing to extract structured epilepsy data from unstructured clinic letters: development and validation of the ExECT (extraction of epilepsy clinical text) system," *BMJ Open*, vol. 9, no. 4, e023232, 2019, doi: 10.1136/bmjopen-2018-023232.

[2] B. Fonferko-Shadrach *et al.*, "Annotation of epilepsy clinic letters for natural language processing," *Journal of Biomedical Semantics*, vol. 15, no. 17, 2024, doi: 10.1186/s13326-024-00316-z.

[3] K. Xie *et al.*, "Extracting seizure frequency from epilepsy clinic notes: a machine reading approach to natural language processing," *Journal of the American Medical Informatics Association*, vol. 29, no. 5, pp. 873–881, 2022, doi: 10.1093/jamia/ocac018.

[4] R. Abeysinghe, S. Tao, S. D. Lhatoo, G.-Q. Zhang, and L. Cui, "Leveraging pretrained language models for seizure frequency extraction from epilepsy evaluation reports," *npj Digital Medicine*, 2025, doi: 10.1038/s41746-025-01592-4.

[5] B. Holgate, J. Davies, S. Fang, J. Winston, J. Teo, and M. Richardson, "Fine-tuning LLMs to extract epilepsy seizure frequency data from health records," in *Proc. 24th Workshop on Biomedical Language Processing*, Vienna, Austria, 2025, pp. 44–55, doi: 10.18653/v1/2025.bionlp-1.5.

[6] Y. Gan, S. H. Barlow, B. Holgate, J. Davies, J. T. Teo, J. S. Winston, and M. P. Richardson, "Reproducible synthetic clinical letters for seizure frequency information extraction," arXiv: 2603.11407, 2026.